

Student Id:

Signature:

09.11.2022

Name:

You have 70 minutes. Good luck!

CSE 114 – Fundamentals of Computer Programming Midterm (2022 Fall)

1) a) (15 p) Write a function *int unique(int n)* which receives an integer *n* such that $9 < n < 1000$ and returns 1 if all its digits are different; 0 otherwise.

b) (15 p) Write the main function which reads a positive integer *k* (assume that $50 \leq k \leq 950$) from the user and prints the first 25 numbers larger than *k* with unique digits.

For example, for *k* = 50 you should print: 51 52 53 54 **56** 57 58 59 60 ...

for *k* = 955 you should print: 956 957 958 **960** 961 962 ...

2) a) (10 p) Using **while loops**, write a function *void f(int a)* which prints the following.

a = 3

3 3 3

a = 5

5 5 5 5 5

a = 9

9 9 9 9 9 9 9 9

b) (20 p) Using **for loops** and the function *f*, write a function *void g(int a, int b)* which prints the following.

a = 4 b = 7

4 - 4 4 4 4

5 - 5 5 5 5 5

6 - 6 6 6 6 6 6

7 - 7 7 7 7 7 7 7

a = 2 b = 3

2 - 2 2

3 - 3 3 3

you can assume that

 $0 < a < b$

3) a) (20 p) Write the following function which encodes an array *A* into another array *Encoded* as in the example. Assume all positive numbers except a -1 marking the end of the values.

A:	4	4	4	4	4	4	7	7	11	11	11	11	-1			
Encoded:	4	6	7	2	11	4	-1									

```
void encode( int A[], int Encoded[] ){
```

b) (20 p) Write the following function which decodes an array *Encoded* into another array *A* as in the example. Assume all positive numbers except a -1 marking the end of the values.

Encoded:	4	1	8	3	25	4	6	2	-1							
A:	4	8	8	8	25	25	25	25	6	6	-1					

```
void decode( int Encoded[], int A[] ){
```